

VentureMiner AI

DevOps & Infrastructure

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Document 23 — DevOps & Infrastructure

How the platform is built, deployed, and operated day-to-day. IaC, environments, release flow, and operational tooling.

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1. Purpose & Scope

This document is the contract for DevOps and infrastructure. It governs how code becomes a running service, how environments are managed, and how the platform is operated.

2. Principles

1. Everything as code. Infrastructure, configuration, runbooks.
2. Boring is good. We pick mature, well-supported tools.
3. Reversible. Every change has a defined rollback.
4. Observable by default. Every service emits traces, logs, metrics from day one.
5. Cost-aware. Engineers see the cost of what they ship.

3. Infrastructure as Code

- Tool: Terraform.
- State: S3 + DynamoDB lock; per environment.
- Modules: a vma- prefixed library; services consume via modules.

4. Environments

dev	local	docker-compose	synthetic
ci	ephemeral per PR	GitHub Actions + Terraform	synthetic
staging	pre-prod	Terraform	anonymized prod clone
canary	production-like	Terraform	live mirror (read-only)
production	live	Terraform	customer

Each environment is in its own AWS account; cross-account access is least-privilege.

5. Source control & branching

- Monorepo at github.com/ventureminer.
- Branching: trunk-based with short-lived feature branches.
- Branch naming: /- (e.g. feat/VM-142-opportunity-bulk-actions).
- Commits: Conventional Commits.
- PRs: small (< 400 lines), 2+ reviewers for sensitive areas, CODEOWNERS enforced.

6. Build & artifact

- Build: GitHub Actions; matrix per service.
- Artifact: OCI images; pushed to ECR with content-addressable tags.
- SBOM: generated per build; signed (Sigstore).
- Vulnerability scan: Snyk per build; failure on critical.

7. Release flow

PR → CI (lint, unit, integration, contract) → main → staging deploy → smoke → tag → canary deploy → soak → prod deploy

- PR: required checks pass; CODEOWNERS satisfied.
- Main: every merge auto-deploys to staging.
- Tag: weekly release tag; auto-deploys to canary.
- Soak: canary holds for 24h with SLO monitoring.
- Prod: auto-promote if SLOs hold; otherwise hold for human review.

8. Deployment

- Strategy: rolling for stateless; blue/green for AI plane and DB.

9. Feature flags

- Tool: LaunchDarkly (managed).
- Default: off.
- Naming: ..
- Lifetimes: flags older than 90 days are removed or graduated.
- Audit: every flag change is logged.

10. Database operations

- Migrations: Atlas; PR-based; CI on ephemeral DB.
- Backups: daily full + continuous WAL; PITR to 35 days.
- Restore drills: quarterly.
- Schema change discipline (Document 05 §10).
- Long-running migrations: scheduled window, expand-contract, monitored.

11. Cost engineering

- Per-service cost dashboards.
- Tagging: every resource tagged with service, env, owner.
- Budgets: per service; alerts at 80%, 100%.
- Right-sizing: monthly review.
- Anomaly detection: cost spike alert.

12. Operational tooling

- Observability: Datadog (logs, metrics, traces, RUM, synthetics).
- Incident: PagerDuty.
- On-call: PagerDuty schedule; weekly rotation.
- Status page: public; auto-updated from health checks.
- Runbooks: in docs/runbooks/, owned by service teams.
- ChatOps: Slack-integrated deploys, rollbacks, feature-flag toggles.

13. Disaster recovery

- RPO: 1h.
- RTO: 4h.
- DR region: us-west-2 (warm standby).

14. Appendix

14.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

14.2 Cross-references

- TRD: Document 02.
- Backend: Document 05.
- Security: Document 21.
- Operations: Document 28.

End of Document 23 — DevOps & Infrastructure. Boring infrastructure is good infrastructure.